IE University
Master in Business Analytics & Data Science

MASTER'S THESIS

Mean, Variance, and Market Efficiency in the Silver Market

Forecasting Weekly Returns and Realised Volatility with Classical, Machine-Learning, and Deep-Learning Models

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Abstract

This thesis asks how much of silver's short-term price behaviour can be forecast, and frames the question through the efficient-market hypothesis (EMH). Silver is modelled at the weekly (Friday-to-Friday) horizon over 2015–2026, and the analysis is factorised into the two moments: the conditional *mean* of returns and the conditional *variance*—efficiency constrains the former but not the latter. For the mean, a deliberate ladder of models of increasing flexibility—ARIMA/ARIMAX, VAR, mixed-data-sampling (MIDAS) regressions, random forests, gradient boosting, and an LSTM—is tested against a common random-walk-with-drift benchmark, expanding the information set from silver's own past (weak-form EMH) to a broad set of public information (semi-strong-form EMH): cross-asset returns, silver futures positioning, macroeconomic releases, technical indicators, and sentiment from Reddit and financial news. For the variance, weekly realised volatility is modelled with GARCH and the heterogeneous autoregressive (HAR) model, and the question of interest is whether sentiment improves the forecast. The central finding is a single coherent statement: no model beats the drift benchmark for the conditional mean by a significant margin—weak- and semi-strong-form efficiency hold for weekly silver returns—yet the conditional variance is strongly and significantly forecastable, with sentiment adding a small but robust increment. Predictable volatility alongside an unpredictable mean is exactly what the EMH permits, so the two results are complementary rather than contradictory.